

Gold Layer

Dimensional data warehouse schema reference

Purpose

This catalog documents the Gold layer tables used for analytics and reporting. It describes the business purpose, columns, data types and intended relationships for customer, product and sales data.

OBJECT	TABLE	ROLE
Customer dimension	gold.dim_customers	Customer attributes enriched with demographic and geographic information.
Product dimension	gold.dim_products	Product attributes and classification for analysis.
Sales fact	gold.fact_sales	Transactional sales measures and dates for reporting.

Schema at a glance

gold.dim_customers customer_key (key) Customer attributes	gold.fact_sales customer_key (FK) product_key (FK) Sales measures & dates	gold.dim_products product_key (key) Product attributes
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The fact table references the dimensions using surrogate keys. This follows a common star-schema pattern, with dimensions providing descriptive context for measures in the fact table.

1. Customer Dimension

Table: gold.dim_customers

Purpose: Stores customer details enriched with demographic and geographic data.

Column Name	Data Type	Description
customer_key	INT	Surrogate key uniquely identifying each customer record in the dimension table.
customer_id	INT	Unique numerical identifier assigned to each customer.
customer_number	NVARCHAR(50)	Alphanumeric identifier representing the customer, used for tracking and referencing.
first_name	NVARCHAR(50)	The customer's first name, as recorded in the system.
last_name	NVARCHAR(50)	The customer's last name or family name.
country	NVARCHAR(50)	The country of residence for the customer (e.g., 'Australia').
marital_status	NVARCHAR(50)	The marital status of the customer (e.g., 'Married', 'Single').
gender	NVARCHAR(50)	The gender of the customer (e.g., 'Male', 'Female', 'n/a').
birthdate	DATE	The date of birth of the customer, formatted as YYYY-MM-DD (e.g., 1971-10-06).
create_date	DATE	The date and time when the customer record was created in the system.

2. Product Dimension

Table: gold.dim_products

Purpose: Provides information about products and their attributes.

Column Name	Data Type	Description
product_key	INT	Surrogate key uniquely identifying each product record in the product dimension table.
product_id	INT	A unique identifier assigned to the product for internal tracking and referencing.
product_number	NVARCHAR(50)	A structured alphanumeric code representing the product, often used for categorization or inventory.
product_name	NVARCHAR(50)	Descriptive name of the product, including key details such as type, color, and size.
category_id	NVARCHAR(50)	A unique identifier for the product's category, linking to its high-level classification.
category	NVARCHAR(50)	The broader classification of a product (e.g., Bikes, Components) to group related items.
subcategory	NVARCHAR(50)	A more detailed classification of the product within the category, such as product type.
maintenance_required	NVARCHAR(50)	Indicates whether the product requires maintenance (e.g., 'Yes', 'No').
cost	INT	The cost or base price of the product, measured in monetary units.
product_line	NVARCHAR(50)	The specific product line or series to which the product belongs (e.g., Road, Mountain).
start_date	DATE	The date when the product became available for sale or use.

3. Sales Fact

Table: gold.fact_sales

Purpose: Stores transactional sales data for analytical purposes.

Column Name	Data Type	Description
order_number	NVARCHAR(50)	A unique alphanumeric identifier for each sales order (e.g., 'SO54496').
product_key	INT	Surrogate key linking the order line to the product dimension table.
customer_key	INT	Surrogate key linking the order line to the customer dimension table.
order_date	DATE	The date when the order was placed.
shipping_date	DATE	The date when the order was shipped to the customer.
due_date	DATE	The date when the order payment was due.
sales_amount	INT	The total monetary value of the sale for the line item, in whole currency units (e.g., 25).
quantity	INT	The number of units of the product ordered for the line item (e.g., 1).
price	INT	The price per unit of the product for the line item, in whole currency units (e.g., 25).

Relationships and modeling notes

- **Dimension joins:** Join gold.fact_sales.customer_key to gold.dim_customers.customer_key, and gold.fact_sales.product_key to gold.dim_products.product_key.
- **Fact grain:** Confirm and document whether one row represents one order line (one product on an order) or another level of detail. If orders can contain multiple products, order_number alone should not be treated as a unique row key.
- **Surrogate keys:** customer_key and product_key are described as surrogate keys. Define them as primary keys in the dimension tables and foreign keys in the fact table if that matches the implemented database.
- **Date semantics:** order_date, shipping_date and due_date represent separate business events. A dedicated date dimension may be useful for richer calendar analysis.
- **Data type check:** create_date is described as a date and time, but its declared type is DATE. Use DATETIME/TIMESTAMP if time-of-day must be retained. Currency fields should document currency and precision; DECIMAL may be appropriate where fractional values are possible.
- **Product history:** If product attributes change over time, define whether the dimension stores only current values or tracks historical versions (for example, using a Type 2 slowly changing dimension).

Catalog scope: This document reflects the supplied data dictionary. Primary keys, foreign keys, nullability, uniqueness, currency, source systems and refresh cadence should be verified against the implemented warehouse before being treated as enforced constraints.